

Trade Policy and Welfare

Strategic Policy Interactions

Lecture Notes

Roadmap

These notes cover the welfare economics of trade policy in five stages: (I) partial-equilibrium effects of tariffs, export subsidies, and quotas in small and large countries; (II) the theory of the second best when domestic or foreign market power is present, developed through Cournot duopoly and reaction-function (best-response) analysis; (III) the classic theory of *optimum tariffs and retaliation*, tying the classroom Cournot model to Johnson (1953–54), Mayer (1981), and Syropoulos (2002); (IV) strategic trade and industrial policy under imperfect competition, following Brander and Spencer (1984, 1985) from the foreign-monopoly tariff problem through reciprocal dumping to strategic export subsidies, with a Cournot-versus-Bertrand caveat on when the optimal policy flips from a subsidy to a tax; and (V) the political economy of trade policy, where Mayer (1984) and Grossman and Helpman (1994) explain why actual protection deviates from these welfare-maximizing benchmarks. A worked problem set (with an original and a modified, solved question) appears at the end.

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1 Trade Policy Instruments: An Overview

- **Trade Policies**

- Tariffs
- Customs (non-tariff administrative barriers)
- Export subsidies

- **Optimal Policies** — policy design that maximizes national (or, if cooperative, joint) welfare.

- **Strategic Interactions**

- Tariff wars (non-cooperative retaliation)
- Quotas

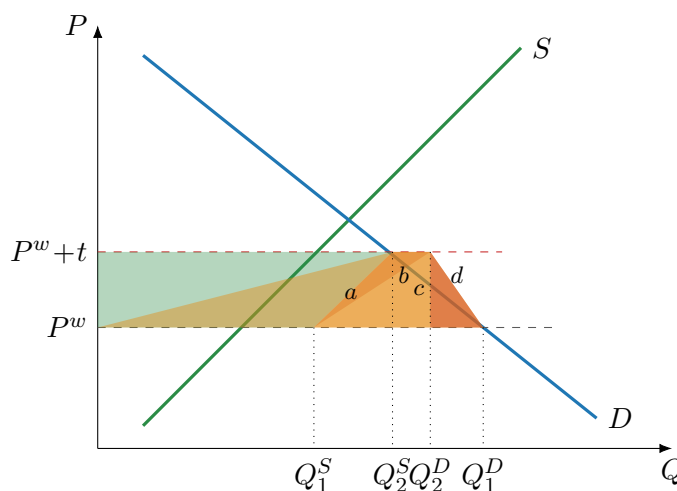
- **Trade Agreements** — cooperative resolutions of the strategic tariff-setting game.

This is the organizing logic of the whole course: we move from a single country taking the rest of the world as given (Part I), to a single country with *market power* interacting with foreign competitors who react optimally (Part II), and finally to the fully strategic, game-theoretic problem of *two* policy-setting governments (Part III).

2 Partial-Equilibrium Effects of a Tariff

2.1 The Small-Country Case: Consumer Surplus, Producer Surplus, and Revenue

A country is “small” if it faces a perfectly elastic (horizontal) world supply/demand for the imported good at price P^w . Let $D(P)$ and $S(P)$ be domestic demand and supply, and let t be a specific tariff so that the domestic price becomes $P_t = P^w + t$.



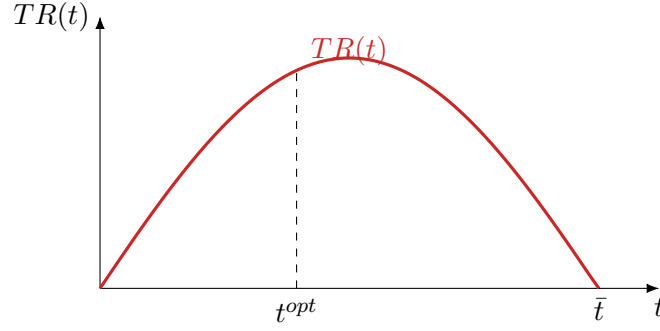
Welfare decomposition (small country)

$$\Delta CS = -(a + b + c + d), \quad \Delta PS = +a, \quad \Delta TR = +c,$$

$$\boxed{\Delta W = \Delta CS + \Delta PS + \Delta TR = -(b + d) < 0.}$$

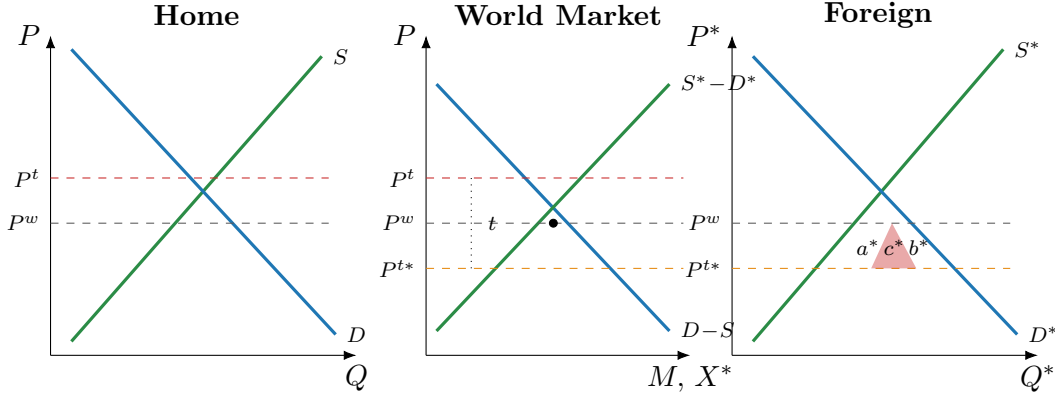
A tariff in a small country is unambiguously welfare-reducing: it only creates deadweight loss (regions b and d) because the country cannot improve its terms of trade — the world price P^w is fixed.

Tariff revenue as a function of the tariff rate traces out a “Laffer curve”: it is zero at $t = 0$, rises, reaches a maximum, and returns to zero at the prohibitive tariff $\bar{t}$ (the rate that eliminates trade entirely).



2.2 The Large-Country Case: Terms-of-Trade Gains

If the country is “large”, its tariff drives a wedge between the price paid domestically, P^t , and the (lower) price received by foreign exporters, P^{t*} , with $P^t - P^{t*} = t$. The following three-panel diagram — Home market, world market (excess demand $D-S$ meeting foreign excess supply S^*-D^*), and Foreign market — is the standard device for tracking this.



Large-country tariff decomposition

Relative to free trade at P^w : Home consumers face a higher price $P^t = P^w + t$ (losing surplus), Home producers gain, the government collects revenue $t \times M$ on the new import volume M , and — crucially — the price *received* by foreign exporters falls to $P^{t*} < P^w$. That fall is a terms-of-trade gain captured in Home’s tariff revenue and is a pure transfer from Foreign (areas a^*, c^*, b^* in the Foreign panel measure the corresponding loss in Foreign surplus). Formally, with $e \equiv (P^w - P^{t*}) M$ the terms-of-trade gain,

$$\Delta W_{\text{Home}} = -(b + d) + e,$$

which is *positive* for a sufficiently small t — this is the classical **optimum tariff argument** (Bickerdike–Kaldor–Johnson).

2.3 The Optimal Tariff Formula

Let ε_2^* denote the (compensated) price elasticity of the *foreign export supply* curve (equivalently, of foreign’s offer curve / reciprocal demand) evaluated at the equilibrium. Maximizing Home

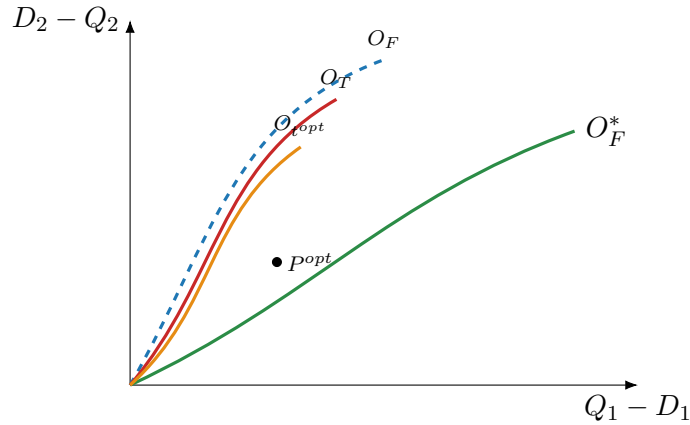
welfare $E_u du = -M_2 d\pi + \pi dM_2$ (envelope-theorem condition on Home's expenditure function) with respect to the tariff yields:

Johnson's optimum tariff formula

$$t^{opt} = \frac{1}{\varepsilon_2^*}, \quad \text{where} \quad \varepsilon_2^* = \xi_1^* - 1,$$

with ξ_1^* the elasticity of Home's demand for imports as reflected in Foreign's offer curve. The optimal tariff is *inversely* related to the elasticity of the trade partner's offer curve: the less elastic Foreign's willingness to supply exports, the higher is Home's optimal (terms-of-trade-improving) tariff. If Foreign's offer curve is a straight line through the origin (infinitely elastic in the relevant sense, i.e. the small-country/perfectly competitive benchmark), $\varepsilon_2^* \rightarrow \infty$ and $t^{opt} \rightarrow 0$: a small country's optimal unilateral tariff is zero.

This result is most transparent in **offer-curve** (reciprocal-demand) space. Let O_F^* denote Foreign's free-trade offer curve, O_T Home's tariff-distorted offer curve for an arbitrary tariff, and $O_{t^{opt}}$ Home's offer curve under the optimal tariff — the one tangent to Home's highest attainable community indifference curve given O_F^* .



Each successively steeper Home offer curve ($O_F \rightarrow O_T \rightarrow O_{t^{opt}}$) corresponds to a higher tariff and intersects O_F^* at a point further from the free-trade point along Foreign's offer curve; the optimal tariff is the one whose intersection point lies on Home's *highest* attainable indifference curve — i.e. where a Home indifference curve is tangent to O_F^* .

2.4 The Tariff (Standard Small-Open-Economy Case, Diagrammatic Recap)

Specific tax on imports t : $P_t = P_t^* + t$ and equilibrium requires

$$MD(P_t) = XS^*(P_t^*),$$

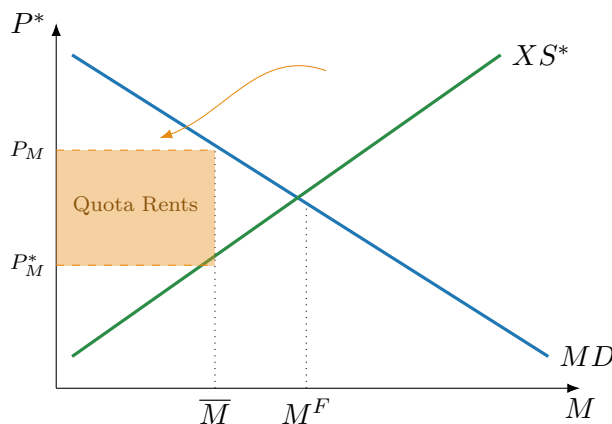
i.e. import demand at the tariff-inclusive price equals foreign export supply at the tariff-exclusive (received) price.

$$\Delta CS = -(a + b + c + d), \quad \Delta PS = +a, \quad \Delta TR = +(c + e), \quad \boxed{\Delta W = -b - d + e}$$

where e is the terms-of-trade gain (zero for a small country, positive for a large country — consistent with the two cases above).

2.5 Import Quotas

An import quota $\bar{M}$ restricts the physical quantity of imports rather than taxing them. With foreign export supply XS^* and home import demand MD , a binding quota $\bar{M} < M^F$ (free-trade imports) creates a wedge between the domestic price P_M and the price P_M^* that would clear the residual (unrestricted) foreign supply — the rectangle between these prices over the quota quantity is the **quota rent**.



Tariff-quota equivalence and its breakdown

A tariff t and a quota $\bar{M} = M(P^w + t)$ generate the *same* price and quantity outcomes under perfect competition — the only difference is who captures the rectangle: the government (tariff revenue) versus whoever holds the import license (quota rents, f). If rents are captured domestically and redistributed lump-sum, $\Delta W = -b - d + f$, identical to the tariff case. If licenses are auctioned to (or captured by) foreigners, $\Delta W = -(b + d)$ only — home loses the rectangle entirely. **The equivalence breaks down** once there is domestic market power (Section 3.3): a quota fixes quantity and hands the domestic monopolist full pricing power over the residual demand, which a tariff does not.

3 Domestic Market Power and the Theory of the Second Best

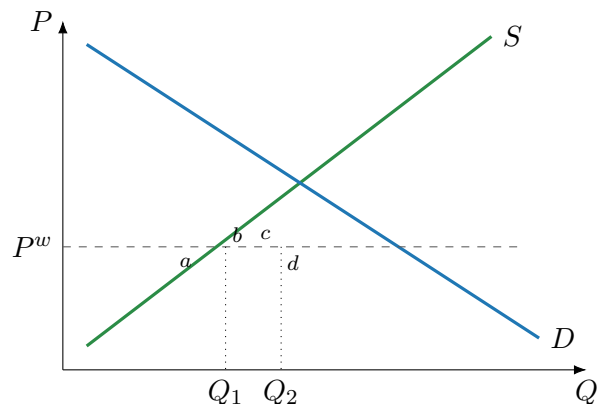
3.1 Introducing a Distortion

- **Theory of the second best in international trade** (introducing a distortion).
- **Domestic market power**: tariffs, quotas.
- **Non-cooperative interaction**: best-response functions, Nash equilibrium, comparative statics.

When a domestic industry itself has market power (e.g. a single domestic producer, or a small number of domestic and/or foreign oligopolists selling into the home market), the “first-best” free-trade prescription no longer applies: a well-chosen trade instrument can act as the *second-best* correction for the domestic distortion (market power) even absent any terms-of-trade motive.

3.2 Duopoly with Homogeneous Goods — the Standard Tariff Case Reprised

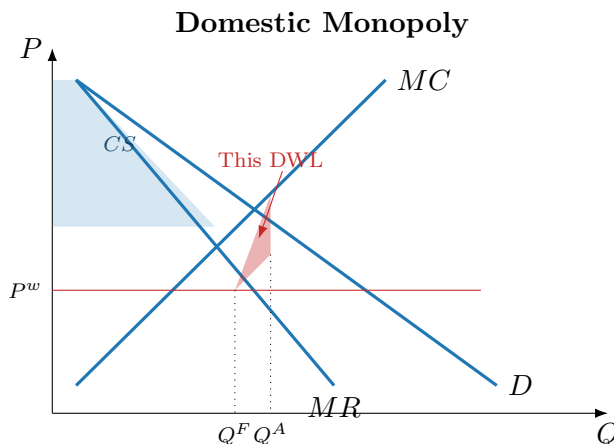
Two firms, home and foreign, compete in *quantities* (Cournot) selling a homogeneous good into the home market. With no domestic market power beyond the standard partial-equilibrium picture, a home tariff still produces the same qualitative decomposition as in Section 2:



$$\Delta W = -b - d \quad (\text{"Tariffs, standard case"})$$

3.3 Domestic Monopoly Protected by a Tariff

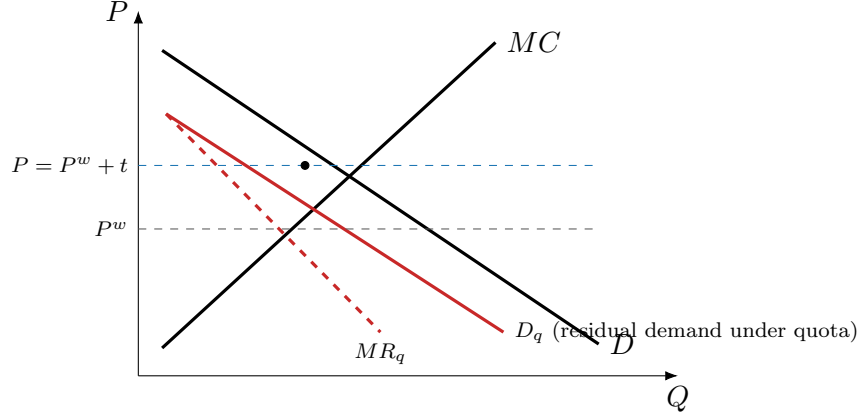
Now suppose the home industry is a *single* domestic firm shielded from perfectly elastic foreign supply at P^w . Under free trade the firm cannot price above P^w (it would be undercut by imports), so it behaves as a price-taker: $P = P^w$, output Q^F where $MC(Q^F) = P^w$.



Tariff vs. prohibitive protection under domestic monopoly

A tariff $t < \bar{t}$ raises the effective import price to $P^w + t$; the monopolist still cannot price above this ceiling, so it remains a (constrained) price-taker at $P = P^w + t$, expanding output along MC up to $Q^A(t)$. Welfare falls by the usual $-(b + d)$ *plus* whatever monopoly distortion remains. Only once the tariff is *prohibitive* ($t \geq \bar{t}$, so that imports fall to zero, $Q^F \rightarrow 0$) does the domestic firm become an unconstrained monopolist, setting $MR = MC$ at a small output and driving price up sharply — the shaded “This DWL” triangle. **A quota is worse:** even a *non-prohibitive* quota $\bar{M}$ hands the monopolist an effectively vertical

(perfectly inelastic) segment of import supply at $\bar{M}$, so the monopolist faces the *residual demand* $D_q(P) = D(P) - \bar{M}$ and its associated MR_q , and optimizes fully — reintroducing the monopoly distortion at *any* binding quota level, however small.



3.4 Best-Response Functions and Nash Equilibrium in a Duopoly

Two firms (or, later, two *governments*) choose actions a_1, a_2 (quantities, or tariffs) to maximize payoffs $\pi^1(a_1, a_2; \theta_1)$ and $\pi^2(a_1, a_2; \theta_2)$, where θ_i is an exogenous shifter (e.g., a cost parameter or a policy variable of a third party).

First- and second-order conditions

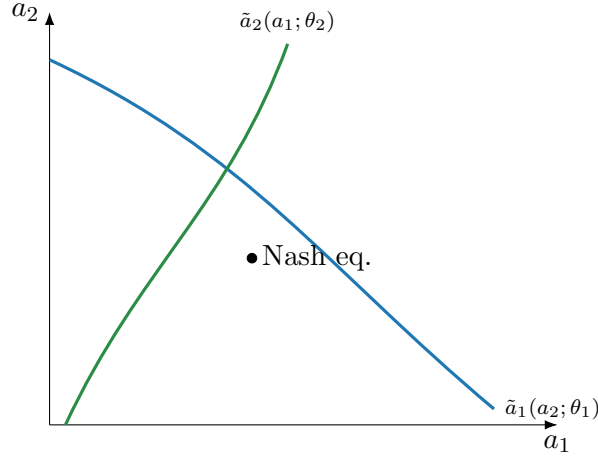
(1) First-order (best response) conditions:

$$\pi_{a_1}^1 = 0 \implies \tilde{a}_1(a_2; \theta_1), \quad \pi_{a_2}^2 = 0 \implies \tilde{a}_2(a_1; \theta_2).$$

(2) Second-order (concavity) conditions:

$$\pi_{a_1 a_1}^1 < 0, \quad \pi_{a_2 a_2}^2 < 0, \quad \Delta \equiv \pi_{a_1 a_1}^1 \pi_{a_2 a_2}^2 - \pi_{a_1 a_2}^1 \pi_{a_2 a_1}^2 > 0.$$

($\Delta > 0$ is the Nash-equilibrium stability/uniqueness condition — the analogue of $\Delta = P'(3P' + P''X) > 0$ used below for the Cournot duopoly, and of Syropoulos's $A \equiv \varepsilon^1 + \varepsilon^2 - 1 > 0$ Marshall–Lerner-type stability condition.)



3.5 Strategic Complements versus Substitutes

Definition

$$\pi_{a_1 a_2}^1 \begin{cases} > 0 & \Rightarrow \text{strategic complements} \\ < 0 & \Rightarrow \text{strategic substitutes} \end{cases}$$

When actions are strategic *substitutes* the best-response curves are negatively sloped (as drawn above): an increase in the rival's action makes one's own marginal payoff *fall*, so one responds by *reducing* one's own action. Under strategic *complements* best-response curves slope up (both actions move together).

3.6 Comparative Statics via Cramer's Rule

Totally differentiate the two first-order conditions with respect to a shock $d\theta_2$ (holding θ_1 fixed):

$$d\pi_{a_1}^1 = 0 \Rightarrow \pi_{a_1 a_1}^1 da_1 + \pi_{a_1 a_2}^1 da_2 = 0,$$

$$d\pi_{a_2}^2 = 0 \Rightarrow \pi_{a_2 a_1}^2 da_1 + \pi_{a_2 a_2}^2 da_2 + \pi_{a_2 \theta_2}^2 d\theta_2 = 0.$$

In matrix form,

$$\begin{pmatrix} da_1 \\ da_2 \end{pmatrix} = \frac{1}{\Delta} \begin{pmatrix} \pi_{a_2 a_2}^2 & -\pi_{a_1 a_2}^1 \\ -\pi_{a_2 a_1}^2 & \pi_{a_1 a_1}^1 \end{pmatrix} \begin{pmatrix} 0 \\ -\pi_{a_2 \theta_2}^2 d\theta_2 \end{pmatrix}.$$

This Cramer's-rule system is the workhorse for *every* comparative-statics result in this course: it tells us how a change in one government's (or firm's) exogenous parameter ripples through both players' equilibrium actions once we account for strategic feedback.

3.7 Envelope Theorem and Welfare (Policy) Comparative Statics

Since firm/government 1 is optimizing over a_1 at the equilibrium, $\pi_{a_1}^1 = 0$, so by the **envelope theorem** the *direct* effect of a_1 on π^1 drops out of any comparative-statics welfare expression:

$$\frac{d\pi^{1N}}{d\theta_2} = \underbrace{\pi_{a_1}^1}_{=0} \frac{da_1^N}{d\theta_2} + \pi_{a_2}^1 \frac{da_2^N}{d\theta_2}, \quad \frac{d\pi^{2N}}{d\theta_2} = \pi_{a_1}^2 \frac{da_1^N}{d\theta_2} + \underbrace{\pi_{a_2}^2}_{=0} \frac{da_2^N}{d\theta_2} + \pi_{\theta_2}^2.$$

Only the terms operating *through the rival's* equilibrium action survive for the firm not directly hit by θ_2 — a preview of the “profit-shifting” logic of Part 3. If θ_2 is interpreted as a specific cost wedge induced by a tariff, $c_2 = c_1 + t$, then national welfare is $U = \pi^W + \pi^*$ evaluated at $c_2 = c_1 + t$: raising t is formally identical to raising firm 2's marginal cost.

4 A Cournot Duopoly Model of Strategic Trade Policy

4.1 Setup

Two firms supply a homogeneous good to the home market: firm 1 (home) with constant marginal cost c_1 and firm 2 (foreign) with constant marginal cost c_2 . Let X_1, X_2 be their outputs, $X = X_1 + X_2$ aggregate supply, and inverse demand $P(X)$ with $P'(X) < 0$. Profits are

$$\pi^1(X_1, X_2) = P(X)X_1 - c_1X_1, \quad \pi^2(X_1, X_2; c_2) = P(X)X_2 - c_2X_2.$$

First- and second-order conditions

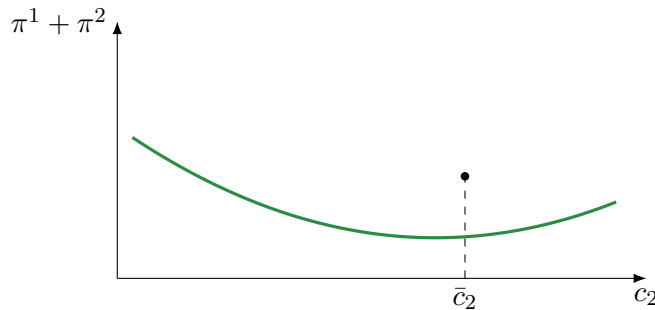
$$\begin{aligned} \pi_{X_i}^1 &= P + P'X_i - c_i = 0, \quad i = 1, 2, \\ \pi_{X_iX_i}^1 &= 2P' + P''X_i < 0 \quad (\text{SOC}), \quad \pi_{X_iX_j}^1 = P' + P''X_i < 0 \quad (\text{strategic substitutes}). \\ \Delta &\equiv \pi_{X_1X_1}^1 \pi_{X_2X_2}^2 - \pi_{X_1X_2}^1 \pi_{X_2X_1}^2 > 0. \end{aligned}$$

4.2 Comparative Statics with Respect to Rival's Cost, c_2

Totally differentiating both FOCs with respect to c_2 (an increase in the *foreign* firm's marginal cost, e.g. via a home tariff, $c_2 = c_1 + t$):

$$\frac{dX_1}{dc_2} = \frac{-\pi_{X_1X_2}^1}{\Delta} = \frac{-(P' + P''X_1)}{\Delta} > 0, \quad \frac{dX_2}{dc_2} = \frac{2P' + P''X_1}{\Delta} < 0, \quad \frac{dX}{dc_2} = \frac{P'}{\Delta} < 0.$$

An increase in the rival's cost raises one's *own* output, lowers the *rival's* output by more, and lowers total output (raising price) — the standard Cournot “cost pass-through” result, and the reason a tariff on a foreign duopolist can *shift* profit toward the home firm.



4.3 Effect on Individual Profits

By the chain rule (using $\pi_{X_2}^1 = P'X_1 < 0$, $\pi_{c_2}^2 = -X_2 < 0$, $\pi_{X_1}^2 = P'X_2 < 0$):

$$\frac{d\pi^1}{dc_2} = \pi_{X_2}^1 \frac{dX_2}{dc_2} = \underbrace{(-)}_{\pi_{X_2}^1} \cdot \underbrace{(-)}_{dX_2/dc_2} > 0,$$

$$\frac{d\pi^2}{dc_2} = \pi_{X_1}^2 \frac{dX_1}{dc_2} + \pi_{c_2}^2 = \underbrace{(-)}_{\pi_{X_1}^2} \cdot \underbrace{(+)}_{dX_1/dc_2} + \underbrace{(-)}_{\pi_{c_2}^2} < 0.$$

So an increase in the rival's cost *always* raises own profit and *always* lowers the rival's profit — but the effect on the *sum* $\pi^1 + \pi^2$ is ambiguous in general (it depends on curvature of $P(\cdot)$), producing the U-shaped joint-profit curve sketched above: near $c_2 = c_1$ a small cost increase can actually *reduce* the joint pie (a pure business-stealing/reallocation effect dominates), while for larger wedges the home-firm gain eventually dominates and joint profit recovers, flattening out once firm 2 is priced out of the market entirely ($X_2 \rightarrow 0$).

4.4 The Symmetric Benchmark, $c_2 = c_1$

At the symmetric point $X_1 = X_2 = X$, one can show

$$\Delta \equiv P'(3P' + P''X) > 0, \quad \left. \frac{d(\pi^1 + \pi^2)}{dX_2} \right|_{c_2=c_1} = X_1 \left[\frac{(P')^2}{\Delta} - 1 \right] < 0.$$

So starting from symmetry, a small *unilateral* increase in firm 2's cost lowers joint industry profit — but, as just noted, this is purely a local (near-symmetric) result; it is *not* the same as the effect on national welfare, which also counts consumer surplus, as we develop next.

4.5 Quasilinear National Welfare

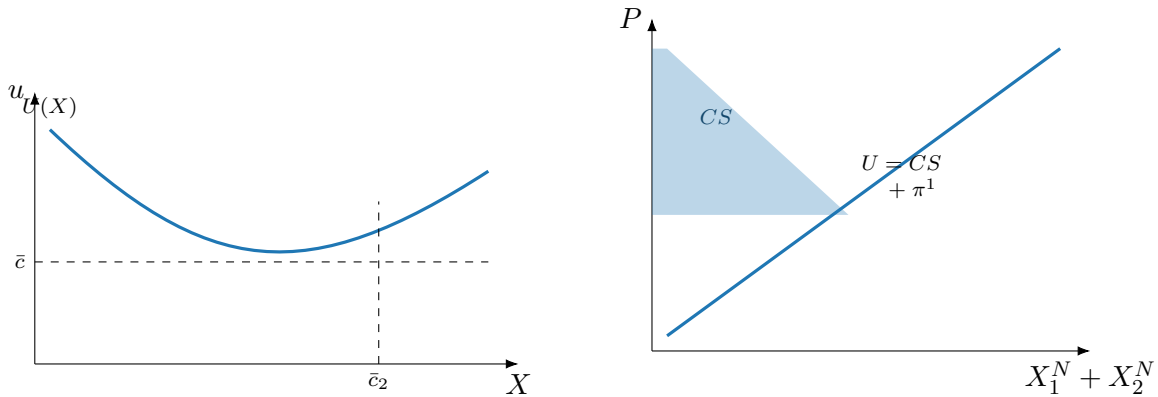
Introduce a numeraire good Z (with price normalized to 1, using labor as the only additional input, wage $w = 1$, so $Z = L_Z$), and let consumers have quasilinear preferences

$$U(X, Z) = u(X) + Z, \quad \text{budget: } PX + Z = Y = wL + \pi^1 + \pi^2,$$

so that $Z = Y - PX$ and

$$U = u(X) - PX + \pi^1 + \pi^2 + wL, \quad \underbrace{u(X) - PX}_{CS} + \underbrace{\pi^1 + \pi^2}_{\Pi}.$$

The consumer's first-order condition is $u'(X) = P$ (inverse demand = marginal utility).



4.6 Welfare Effect of a Small Tariff, dU/dt at $t = 0$

Let $c_2 = c_1 + t$ (the home tariff drives a wedge equal to t between foreign and home marginal cost as perceived in the home market). Differentiate $U = u(X) - PX + \pi^1 + \pi^2$ using $X = X_1 + X_2$, $u'(X) = P$, and the envelope theorem on each firm's optimized profit:

$$\begin{aligned} dU &= u'(X) dX - P dX - X dP + P dX_1 + X_1 dP + P dX_2 + X_2 dP - c_1 dX_1 - c_2 dX_2 - X_2 dt \\ &= (P - c_1) dX_1 + (P - c_2) dX_2 - X_2 dt. \end{aligned}$$

Using $P - c_2 = P - c_1 - t$ and collecting terms,

$$dU = (P - c_2) dX + (c_2 - c_1) dX_1 - X_2 dt = (P - c_2) dX + t dX_1 - X_2 dt.$$

Result: sign of the marginal tariff

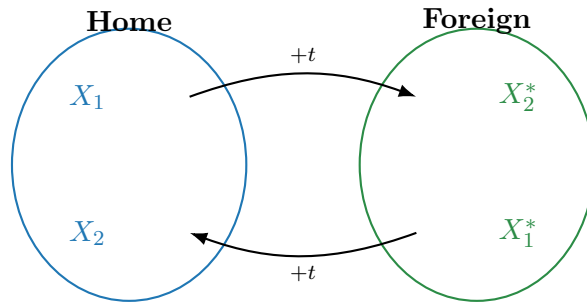
Evaluated at $t = 0$ (so $c_2 = c_1$, hence $P = c_2$ is *not* generally true — rather $P > c_i$ by market power, and the first term vanishes only when using the correct envelope condition on *both* firms' outputs jointly):

$$dU|_{t=0} = (P - c_1 - t) d\tilde{X} - X_2 dt \Big|_{t=0} = -X_2 dt < 0.$$

A marginal home tariff, starting from free trade, is welfare-reducing in this baseline quasilinear, single-market Cournot model: the profit-shifting gain to the home firm is a second-order (envelope) effect, while the consumption loss from a higher price is first order. This mirrors Syropoulos's Proposition 1(vii)–(viii) below: the classic *first-order* optimum-tariff argument requires either (i) genuine terms-of-trade power in a competitive model (Section 2.3), or (ii) a strictly *positive*, non-marginal profit-shifting motive of the Brander–Spencer type once the home firm is not simply a price-taker. When the home market is served *only* by home and foreign duopolists with *no* consumers abroad benefiting from retaliation, a small tariff still reduces domestic welfare on the margin because the term $-X_2 dt$ (lost consumer/rectangle surplus on the foreign firm's units) dominates locally.

4.7 The Symmetric Two-Country (Reciprocal Dumping) Extension

Finally, suppose *both* countries have a home firm and each ships to both markets (“cross-hauling”): Home firm sells X_1 at home and X_2^* abroad; Foreign firm sells X_2 at home and X_1^* abroad, each subject to the trade cost/tariff t (or $\bar{t}$) on the exported leg.



By symmetry, $X_1^* = X_2$ and $X_2^* = X_1$, and Home welfare is

$$U = \underbrace{u(X_1 + X_2) - P_1(X_1 + X_2)}_{CS} + \underbrace{\pi^1(X_1 + X_2)}_{II} + \underbrace{\pi^{1*}(X_1^*, X_2^*, t)}_{\text{home firm's foreign profit}} + wL.$$

This is the natural bridge to *national* (as opposed to purely domestic-market) welfare analysis of strategic trade policy, and to the two-*government* tariff game of Part 4.

5 Optimum Tariffs, Retaliation, and Country Size: Three Classic Papers

This section connects the classroom Cournot/reaction-function machinery above to three papers that form the canonical theory of strategic tariff policy.

5.1 Johnson (1953–54): Optimum Tariffs and Retaliation

Johnson, H.G. (1953–54), Review of Economic Studies

Setting. A two-country, two-good pure-exchange model with well-behaved community indifference curves and offer curves O_I, O_{II} .

Key results.

- Country I's optimal (unilateral) tariff is $t_I = \frac{1}{j-1}$, where j is the elasticity of Country II's reciprocal demand (offer) curve at the post-tariff equilibrium — exactly the $t^{opt} = 1/\varepsilon_2^*$ formula of Section 2.3.
- Because tariff proceeds are redistributed to consumers, imposing t_I rotates Home's offer curve (it does *not* simply shift it outward proportionally); the new curve is tangent to the *same* internal price ratio, tax-inclusive.
- **Retaliation:** Country II's best response to O_I' is its own optimum tariff, $t_{II} = 1/(k-1)$. Iterating this process (a reaction-curve / “welfare-reaction-curve” technique, explicitly likened by Johnson to the *Cournot duopoly solution*) converges either to a **policy (Nash) equilibrium** — tangency of each country's indifference curve with the other's tariff-ridden offer curve — or to a perpetual **tariff cycle**.
- **Central result (contra Scitovsky):** it is *not* true that both countries must lose from retaliation relative to free trade. At most *one* country must lose (since gains require an improved terms of trade that cannot hold for both simultaneously), but either country *can* end up better off under mutual tariffs than under free trade if it has relatively higher/lower demand elasticities as characterized in the constant-elasticity ($X = AY^k$, $Y = BX^j$) example, and even trade volume never falls to zero.
- Whether a country “wins” the tariff war is governed entirely by the relative elasticities j, k of the two reciprocal demand curves — not by any simple, universally rankable index (Johnson explicitly criticizes Kaldor's loose elasticity claim on exactly this point).

5.2 Mayer (1981): Theoretical Considerations on Negotiated Tariff Adjustments

Mayer, W. (1981), Oxford Economic Papers

Setting. Two large countries with utility-in-tariff-space functions $W(t, t^*)$, $W^*(t, t^*)$ derived from the offer-curve diagram exactly as in Johnson.

Key constructs.

- The ***RR locus***: the set of zero-slope points of Home's tariff-indifference curves in (t, t^*) space — i.e., Home's best-response tariff for every possible foreign tariff (Johnson's optimum-tariff reaction curve, restated in tariff rather than offer-curve space). Similarly R^*R^* for Foreign. Their intersection E is the (Nash) **tariff war / conflict equilibrium** — the threat point of any negotiation.
- The ***CC locus***: the Pareto-efficient locus of tangencies between the two countries' tariff-indifference curves. Along CC , domestic relative prices are equalized across countries, which (using $p = \pi(1 + t)$, $p^* = \pi/(1 + t^*)$) implies the Pareto condition

$$t + t^* + tt^* = 0.$$

- **Negotiated settlements**: with an “unconstrained collusive behavior” bargaining rule (each country at least as well off as at the threat point E , no further joint gains possible), the set of feasible agreements is the segment $MN \subset CC$ that Pareto-dominates E . Since E is characterized by *perpendicular* tariff-indifference curves while CC requires *tangency*, some point on CC must dominate E — **negotiated tariffs are always (weakly) better for both countries than a tariff war**.
- **Tariffs by both countries cannot be Pareto efficient**: equation $t + t^* + tt^* = 0$ forces either free trade for both, or a tariff by one paired with a *subsidy* by the other.
- **Free trade need not be attainable through negotiation**: even though the free-trade point lies on CC , it need not lie on the negotiation-relevant segment MN — so bargaining from a tariff-war threat point can settle on a Pareto-efficient outcome that still involves positive tariffs/subsidies.
- In the **large-small country** case, the small country's threat-point tariff is already zero, which further restricts (and can eliminate) the scope for a negotiated move all the way to free trade.

5.3 Syropoulos (2002): How Country Size Matters

Syropoulos, C. (2002), Review of Economic Studies, 69, 707–727

Setting. A general neoclassical two-country, two-good model (GDP and expenditure functions à la Dixit–Norman) in which each country i 's **relative size** is defined as $\lambda^i \equiv L^j/L^i$ ($j \neq i$), holding *per-capita* endowments v^1, v^2 fixed — a cleaner notion of size than GDP shares because it is not endogenous to trade policy.

Key results.

- Each country's Nash-equilibrium tariff satisfies the *same* formula as Johnson's, now written in ad valorem form and general equilibrium:

$$\tau^i = \frac{1}{\varepsilon^i - 1}, \quad \varepsilon^i \equiv \text{own price elasticity of import demand},$$

generating best-response tariff functions $\tau^1(\tau^2)$, $\tau^2(\tau^1)$ whose intersection is the interior Nash tariff equilibrium N .

- **Proposition 1 (limits).** As a trade partner becomes infinitesimally small ($\lambda^j \rightarrow 0$), that partner's Nash and free-trade prices, welfare, and tariff all converge to its *own* free-trade/autarky levels, while the large country's domestic price converges to *its own autarky price* under *both* free trade and retaliation (it becomes a pure price setter, so the terms-of-trade motive for policy vanishes only in this double limit).
- **Proposition 2 (big countries can win).** For each country i there is a threshold relative size $\bar{\lambda}^i \in (0, \infty)$ such that i wins the bilateral tariff war (ends up better off under retaliation than under free trade) whenever its actual relative size exceeds $\bar{\lambda}^i$. Defining $\underline{\lambda}^i \equiv 1/\bar{\lambda}^j$, sizes split into three regions: country j wins ($\lambda^i < \underline{\lambda}^i$), a **Prisoner's Dilemma** region ($\underline{\lambda}^i \leq \lambda^i \leq \bar{\lambda}^i$), and country i wins ($\lambda^i > \bar{\lambda}^i$).
- **Assumptions 1–2 (regularity):** $\theta^i \equiv p^i \varepsilon_{p^i}^i > 0$ (Corden's SOC-type condition) and $\sigma^i > 1$ (elasticity of substitution in consumption exceeds one) jointly imply **Proposition 3:** tariffs are **strategic substitutes** ($\partial \tau^i / \partial \tau^j < 0$), the interior Nash tariff equilibrium is **unique**, and each country's equilibrium tariff is **increasing** in its own relative size and **decreasing** in its partner's — the exact quantity-space analogue of $dX_1/dc_2 > 0$, $dX_2/dc_2 < 0$ from Section 3.2, now derived for tariffs in general equilibrium.
- **Proposition 4** sharpens Proposition 2: under the same regularity conditions the size threshold is unique, and the Prisoner's Dilemma zone is exactly $\underline{\lambda}^i < \lambda^i < \bar{\lambda}^i$ with $\underline{\lambda}^i = 1/\bar{\lambda}^j < 1$.
- A **Heckscher–Ohlin** special case (capital/labor, factor-price equalization under free trade) makes the size $\rightarrow$ elasticity $\rightarrow$ tariff channel fully transparent: raising a country's relative endowment of the factor used intensively in its *import*-competing sector pushes the free-trade world price toward that country's autarky price, illustrating exactly the convergence in Proposition 1(ii).

5.4 Synthesis: How the Three Papers Relate to the Classroom Model

	Johnson (1953–54)	Mayer (1981)	Syropoulos (2002)
Model	2-country exchange, offer curves	2-country tariff-space negotiation	General GE, λ^i size
Optimal tariff	$t = 1/(\varepsilon^* - 1)$	Same, restated via RR locus	$\tau^i = 1/(\varepsilon^i - 1)$
Retaliation	Cournot-style reaction curves; cycles possible	Threat point E (Nash)	Unique Nash under Assump. 1–2
Who wins?	Depends on relative elasticities j, k	N/A (focus on negotiation)	Depends on relative size λ^i vs. threshold
Cooperation	Not modeled	CC locus dominates E ; $t + t^* + tt^* = 0$	Not modeled (single-shot Nash)

In the classroom Cournot model of Section 3, the “elasticity” that determines who gains from a cost/tariff wedge is P' and P'' (curvature of demand); in Johnson and Syropoulos, it is the price elasticity of the *trade partner’s* offer/import-demand curve; in every case the same economic force is at work — **whichever side has the less elastic response bears more of the terms-of-trade / cost burden and the intervening party captures the difference**, subject to Mayer’s caveat that unilateral optimization (retaliation) is always dominated by a negotiated, Pareto-efficient settlement.

6 Trade and Industrial Policies Under Imperfect Competition

Roadmap for this section

1. Domestic market power & protectionism
2. Foreign multinational & tariffs
3. Strategic import policies
4. Strategic export policies
5. Multiple markets

Sections 2–4 above already covered the domestic-market-power case (Section 3.3) and the basic Cournot profit-shifting logic (Section 4). This section completes the roadmap: it revisits the *foreign-monopoly* tariff problem with a general (non-linear) demand curve, develops the full **reciprocal-dumping** model of intra-industry trade with tariffs, and then turns from *import* policy to **export** policy — the Brander–Spencer theory of strategic export subsidies — before closing with the price-versus-quantity-competition caveat that governs whether the optimal export policy is a subsidy or a tax.

6.1 Reciprocal Dumping and Free Entry

Consider first a purely domestic building block. With quasilinear preferences and Q the quantity consumed, indirect welfare is $U = u(Q) - PQ$. Differentiating,

$$dU = (u'(Q) - P) dQ + Q dP.$$

At the consumer's optimum $u'(Q) = P$, so the first term vanishes and $dU = Q dP$. Since $dP = P'(Q) dQ$ along the demand curve, this also gives the useful identity

$$dQ = \frac{dP}{P'(Q)}.$$

Now embed this in a **reciprocal-dumping** model of intra-industry trade (Brander, 1981; Brander and Spencer, 1984, Sec. 4): there are two identical countries, each with one firm, and the two firms compete à la Cournot in *each* national market separately (segmented markets), paying a fixed entry cost f and a per-unit tariff t on whatever they export. Firm i 's profit is

$$\pi^i = \underbrace{P(Q)x_i - cx_i - f}_{\text{home-market sales}} + \underbrace{P(Q^*)x_i^* - (c+t)x_i^*}_{\text{exports, tariff-ridden}},$$

where $Q = x_1 + x_2$ is home consumption and $Q^* = x_1^* + x_2^*$ is foreign consumption. The first-order condition for the export decision is

$$\pi_{x_i^*}^i = 0 \implies P(Q^*) + P'(Q^*)x_i^* - (c+t) = 0.$$

Totally differentiating this FOC and solving for the comparative static of a firm's own export quantity with respect to a perturbation dP (e.g., induced by a change in the rival's cost or the tariff) yields an expression of the form

$$dx_i^* = - \frac{P' + P''x_i^*}{(P')^2} dP.$$

Regularity and the two qualitative cases

Assume, as throughout this course, that $P' + P''x < 0$ (the standard regularity/stability condition used since Section 4.1). Then the sign of dx_i^* is tied to the sign of dP :

$$(1) \ dx > 0 \ \& \ dP > 0, \quad (2) \ dx < 0 \ \& \ dP < 0.$$

Own exports and price move *together*: anything that raises the (net-of-cost) price a firm can command in a market raises its equilibrium sales there, and vice versa. This is the building block used repeatedly below to sign the comparative statics of tariffs and subsidies.

6.2 Foreign Monopoly and the Tariff

Return to a single home market served by a *foreign monopolist* exporting quantity X at marginal cost c , facing a home tariff t . National welfare is

$$U = u(X) - PX + tX + \pi^{\text{domestic}}(x_1, x_2),$$

where the last term (domestic firms' profit) is zero here since there is no domestic producer; we carry it along because the identical expression will have a nonzero domestic-profit term in Section 6.3. The foreign monopolist's profit is

$$\pi(X) = PX - (c + t)X, \quad \text{FOC: } \pi_X = 0 \implies P + P'X - (c + t) = 0,$$

$$\pi_{XX} = 2P' + P''X < 0 \quad (\text{SOC}).$$

Totally differentiating the FOC with respect to t ,

$$d\pi_X = 0 \implies (P' + P''X) dX + P' dX - dt = 0 \implies (2P' + P''X) dX = dt,$$

$$X_t \equiv \frac{dX}{dt} = \frac{1}{2P' + P''X} < 0.$$

Welfare's derivative with respect to the tariff (using the envelope theorem on the monopolist's profit and $u'(X) = P$) is

$$dU = -P'X dX + X dt + t dX, \quad U_t = 0 \implies X_t(t - P'X) + X = 0,$$

$$t^{opt} = X \left(P' - \frac{1}{X_t} \right) = X(P' - MR') = X(1 - 2P' - P''X),$$

where $MR' \equiv 1/X_t = 2P' + P''X$ is the slope of the (inverse) marginal-revenue schedule.

Brander and Spencer (1984), foreign monopoly: the sign of the optimal tariff

Define the **relative convexity of demand**, $R \equiv XP''/P'$. Brander and Spencer (1984) show the optimal tariff can be written compactly as $\hat{t} = -P'X(R + 1)$, and prove:

Proposition (Brander–Spencer). $\hat{t} \geq 0$ as $R \geq -1$.

- *Linear demand* ($P'' = 0 \Rightarrow R = 0$): $\hat{t} = -P'X > 0$ — the familiar positive optimal tariff against a foreign monopolist (a pure **rent-extraction** tariff, since the monopolist earns rents that the tariff can partially capture for the home treasury).
- *Constant-elasticity demand*, $P = X^{-1/\varepsilon}$, $\varepsilon \geq 1$: direct computation gives

$$R = \frac{XP''}{P'} = -1 - \frac{1}{\varepsilon} < -1 \quad \text{for every } \varepsilon > 0,$$

so $\hat{t} < 0$ **always**: the welfare-maximizing policy toward a foreign monopolist facing constant-elasticity demand is an **import subsidy**, not a tariff! The reason is that under a very convex (constant-elasticity) demand curve, a tariff is passed through *more than one-for-one* into the domestic price (the pass-through rate is $p_t \equiv dP/dt = 1/(R + 2)$, which *exceeds* 1 whenever $R < -1$), so the consumer-price cost of extracting monopoly rent swamps the revenue gained.

This is a cautionary result: the standard prescription “tax a foreign monopolist's rents” depends delicately on the curvature of demand, not just on the existence of market power.

6.3 Domestic and Foreign Firms: the Tariff as Profit-Shifting Device

Now reintroduce a *domestic* firm producing y (with fixed cost F and constant marginal cost c , so $\pi(y) = yP(X) - cy - F$) alongside the foreign firm's imports x (profit $V(x, y; t) = xP(X) - kx - tx$, $X = x + y$). Both firms compete à la Cournot in the home market, and the government redistributes tariff revenue lump-sum. National welfare is

$$G(t) = u(X) - PX + tX + \pi(y).$$

The two reaction functions come from $\pi_y = 0$ and $V_x = 0$; stability requires the by-now-familiar determinant condition $D \equiv \pi_{yy}V_{xx} - \pi_{yx}V_{xy} > 0$. Totally differentiating both FOCs with respect to t delivers the comparative statics

$$y_t > 0, \quad x_t < 0, \quad X_t = y_t + x_t < 0.$$

Proposition (Brander and Spencer, 1984): a tariff reduces domestic consumption

Even though total consumption *always falls* ($X_t < 0$) — the home tariff shrinks the market, exactly as in the small-country case of Section 2 — the country typically still gains from the tariff, because it shifts output and profit from the foreign firm to the domestic firm ($y_t > 0$, $x_t < 0$) and collects tariff revenue on the residual imports. Formally, maximizing $G(t)$ over t (setting $G_t = 0$) yields an optimal-tariff condition that nets the standard consumption-side loss $-X_t$ against the profit-shifting/revenue gain x_t ; this is exactly the general-demand, general-cost analogue of the linear-demand Cournot welfare derivative computed in Section 4.6 above, where $dU|_{t=0} = -X_2 dt < 0$ marginally; the worked problem in Section 8.1 below shows that once profit-shifting is allowed to operate *globally* rather than just at the margin, the welfare-maximizing tariff is strictly positive, $t^{opt} = 2\alpha/11$, precisely because profit-shifting eventually dominates the consumption loss.

Brander and Spencer extend this two-firm model to a **two-country** setting in which each country has one domestic and one foreign firm and trade is costly (transport/iceberg cost k), so both markets exhibit two-way (“cross-hauling”) intra-industry trade exactly as in Section 4.8’s Home/Foreign ellipse diagram.

Two countries: transport costs and multilateral liberalization

- **Proposition (transport costs).** If marginal (transport-inclusive) cost k rises, each country’s optimal tariff $\hat{t}$ *falls*. Intuitively, higher transport costs already choke off intra-industry trade, so there is less rent left for a tariff to extract or shift, and the two countries’ markets become more segmented (closer to autarky), reducing each government’s unilateral incentive to intervene.
- **Proposition (jointly excessive protection).** At the noncooperative (each-country-optimal-tariff) equilibrium, *world* welfare would rise if both countries’ tariffs were reduced below their unilaterally optimal levels. Non-cooperatively chosen tariffs exceed the world-welfare-maximizing tariffs — the same qualitative message as Mayer’s (1981) result (Section 5.2) that a negotiated settlement always dominates the tariff-war threat point.

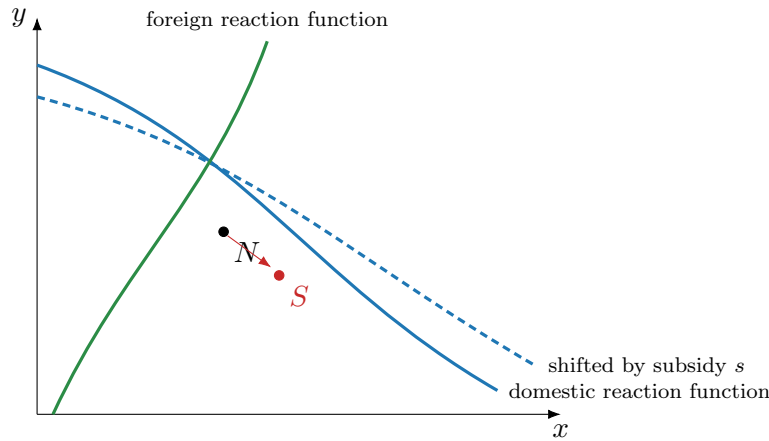
6.4 Strategic Export Policy: Brander and Spencer (1985)

We now turn the analysis around: instead of a government taxing *imports* into its own market, consider a government **subsidizing exports** to a third, purely importing country. Two firms — domestic (output x) and foreign (output y) — compete Cournot for sales in that third market. The domestic firm's variable profit is

$$\pi(x, y; s) = xP(x + y) - c(x) + sx, \quad s = \text{per-unit export subsidy,}$$

$$\pi_x = xP' + P - c_x + s = 0 \quad (\text{FOC}), \quad \pi_{xx} = 2P' + xP'' - c_{xx} < 0 \quad (\text{SOC}),$$

and symmetrically for the foreign firm's profit $\pi^*(x, y) = yP(x + y) - c^*(y)$ with $\pi_y^* = yP' + P - c_y^* = 0$. The regularity conditions $\pi_{xy} = P' + xP'' < 0$, $\pi_{yx}^* = P' + yP'' < 0$ (own marginal revenue falls in rival output — downward-sloping reaction functions) and $D \equiv \pi_{xx}\pi_{yy}^* - \pi_{xy}\pi_{yx}^* > 0$ (stability/uniqueness) are assumed throughout, exactly as in Section 4.1's $\Delta > 0$ condition.



Totally differentiating the two FOCs with respect to s (using $\pi_{xs} = 1$, $\pi_{ys}^* = 0$) and solving by Cramer's rule:

$$x_s \equiv \frac{dx}{ds} = \frac{-\pi_{yy}^*}{D} > 0, \quad y_s \equiv \frac{dy}{ds} = \frac{\pi_{yx}^*}{D} < 0.$$

Proposition 1 (Brander and Spencer, 1985)

An increase in the domestic export subsidy (i) lowers the world price, $p_s = P'(x_s + y_s) < 0$; (ii) raises domestic profit, $\pi_s = xP'y_s + x > 0$ (since $\pi_x = 0$, $\pi_s = x$ by the envelope theorem and $y_s < 0$); and (iii) lowers foreign profit, $\pi_s^* = yP'x_s < 0$.

Domestic welfare. With all output exported (no home consumption yet), domestic surplus net of the subsidy cost is $G(s) = \pi(x, y; s) - sx$. Differentiating and using the envelope theorem ($\pi_s = x$):

$$G_s = xP'y_s - s x_s.$$

At $s = 0$, $G_s = xP'y_s > 0$ (since $P' < 0$, $y_s < 0$): **a marginal export subsidy always raises domestic welfare** starting from free trade — the mirror image of the tariff result in Section 4.6, but with the opposite first-order sign because here *all* output is exported (there is no domestic consumer loss to offset the profit-shifting gain). Setting $G_s = 0$:

$$s^{opt} = \frac{xP'y_s}{x_s} > 0.$$

Proposition 3 (Brander and Spencer, 1985): the Stackelberg-leadership interpretation

The optimal export subsidy moves the industry equilibrium to exactly the point that would prevail, absent any subsidy, if the domestic firm were a **Stackelberg quantity leader**. *Proof sketch.* If the domestic firm led in quantities, its FOC would be $\pi_x(x, y; 0) + \pi_y(x, y; 0) \frac{dy}{dx} = 0$, where $dy/dx = y_s/x_s$ is the slope of the *foreign* firm's reaction function (from totally differentiating $\pi_y^* = 0$). Substituting $dy/dx = y_s/x_s$ and rearranging reproduces exactly condition $\pi_x(x, y; s^{opt}) = 0$ once $s^{opt} = xP'y_s/x_s$ is substituted in — i.e., the subsidy is chosen precisely so as to make the Cournot FOC coincide with the Stackelberg-leader FOC. Diagrammatically (see the reaction-function figure above), the subsidy commits the domestic firm to a more aggressive (outward-shifted) reaction function, moving the Cournot–Nash point from N to S : more domestic output, less foreign output, and a higher domestic profit net of the subsidy cost.

This result is strictly about rivalry. If the domestic firm were an unconstrained *international monopolist* (no rival y), the envelope theorem gives $\pi_s = x$ exactly as before, but now $G_s = -s x_s < 0$ for any $s > 0$: with no rival to shift output away from, an export subsidy is purely wasteful. **Strategic export subsidies are rationalized only by the presence of foreign rivalry** in an imperfectly competitive third market — precisely condition (3)/(4) in the roadmap above (“strategic export policies”).

6.4.1 General Equilibrium and the Terms of Trade

Embedding this in a two-good general-equilibrium economy with utility $u(z, m) = U(z) + m$ (z = consumption of the imperfectly competitive good, m = numeraire), balanced trade, and marginal rate of transformation c_x , the change in real income from a subsidy is

$$d\mathcal{I} = (x - z) dp + (p - c_x) dx,$$

which under pure competition ($p = c_x$) would make $d\mathcal{I} = (x - z) dp < 0$ for any subsidy (worsened terms of trade, no offsetting profit gain) — exactly why export subsidies make no sense in a competitive, two-good neoclassical world. Under Cournot rivalry, however, $p - c_x = -xP' - s$, so

$$\frac{d\mathcal{I}}{ds} = -z p_s + xP'y_s - s x_s, \quad s^{opt} = \frac{xP'y_s}{x_s} - \frac{z p_s}{x_s} > 0.$$

The terms of trade still move against the subsidizing country ($p_s < 0$), but profit shifting (and, if $z > 0$, the extra Pareto gain from expanding under-consumed domestic sales of an imperfectly competitive good) more than compensates.

6.4.2 Two and Three Governments: Nash Subsidy (and Tariff) Equilibria

Propositions 4–5 (Brander and Spencer, 1985): the subsidy war is a Prisoner's Dilemma

If *both* exporting countries subsidize, the Nash equilibrium subsidies are

$$s = \frac{xP'y_s}{x_s} - \frac{zp_s}{x_s} > 0, \quad s^* = \frac{yP'x_{s^*}}{y_{s^*}} - \frac{z^*p_{s^*}}{y_{s^*}} > 0,$$

i.e., **both** countries subsidize in equilibrium (Proposition 4) — yet **joint** welfare of the two exporting countries would *rise* if both subsidies were reduced below their Nash levels (Proposition 5): the subsidy competition is jointly suboptimal, a Prisoner's Dilemma exactly analogous to the tariff-war Prisoner's Dilemma of Syropoulos (2002, Section 5.3) and to Brander–Spencer's own transport-cost Proposition above. If the two exporting nations consumed none of the good themselves, the *jointly* optimal policy would in fact be an export **tax** (not a subsidy) on both firms, sufficient to replicate a two-firm monopoly cartel against the importing country — but this cooperative outcome is not self-enforcing.

Adding a third government — the *importing* country, which sets its own optimal tariff t on total imports $X = x + y$ — closes the model. In the linear-demand, no-home-consumption special case, the three governments' Nash policies satisfy the strikingly clean relationship

$$s = -\frac{1}{2}xP', \quad s^* = -\frac{1}{2}yP', \quad t = -\frac{1}{2}XP' = s + s^* :$$

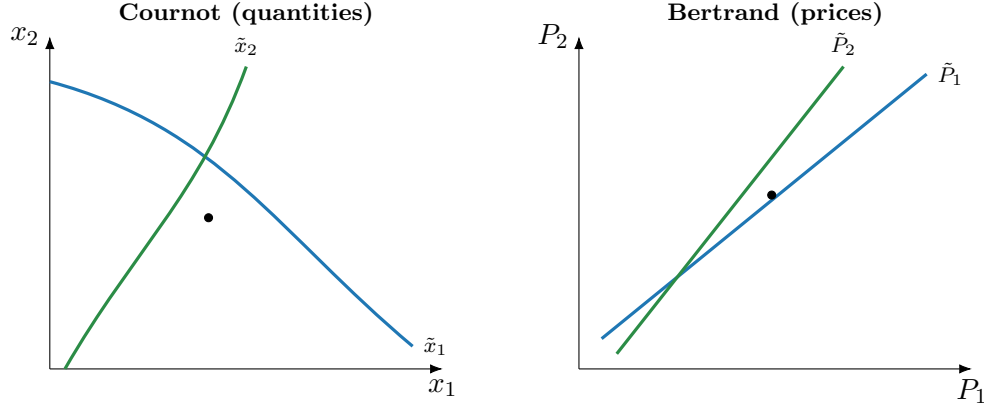
the **importing country's optimal tariff exactly equals the sum of the two exporting countries' optimal subsidies** in this benchmark case, and each exporter's subsidy *raises* the optimal import tariff (the importer captures part of the subsidy-induced rents), while a higher import tariff *reduces* each exporter's optimal subsidy (less rent left to fight over) — a fully closed strategic-trade-policy triangle.

6.5 A Cautionary Coda: Cournot versus Bertrand Competition

Everything above assumed firms compete in **quantities** (Cournot), which is what makes reaction functions downward sloping (**strategic substitutes**, Section 3.5) and makes an *outward* shift of one's own reaction function profitable — hence a subsidy (which shifts the domestic reaction function out) helps the domestic firm by inducing the rival to *contract*. If instead firms compete in **prices** for a differentiated product,

$$\pi(P_1, P_2) = P_1 x_1(P_1, P_2) - (c - s) x_1(P_1, P_2),$$

reaction functions in price space are generally **upward sloping (strategic complements)**: a lower price by firm 1 induces firm 2 to *also* cut its price, rather than raise output as in the Cournot case.



Why the sign of optimal policy can flip

Under price competition, an *export tax* (which raises the domestic firm's effective cost and hence its optimal price) shifts the domestic price-reaction curve *up*; since reaction curves slope up, the rival optimally *raises its price too*, softening competition and raising *both* firms' margins, including the domestic firm's, net of the tax. A subsidy would do the reverse — triggering a price war that dissipates rents. Whether the profit-shifting argument for intervention calls for a **subsidy** (Cournot, strategic substitutes) or a **tax** (Bertrand, strategic complements) therefore hinges entirely on the *mode of competition* and the sign of $\pi_{a_1 a_2}$ from Section 3.5 — a sharp warning against applying the Brander–Spencer subsidy prescription without first verifying how firms actually compete.

7 Rent-Seeking and the Political Economy of Trade Policy

Sections 2–6 all derived trade policy from a benevolent, welfare-maximizing government. This final section asks the opposite question: if governments instead respond to *political* pressure — voters, lobbies, campaign contributions — what tariff structure emerges, and how does it compare to the welfare-maximizing benchmarks derived above? Two complementary answers are given by Mayer (1984), who models tariffs as the outcome of **majority voting** by factor owners, and by Grossman and Helpman (1994), who model tariffs as the outcome of **lobbying and campaign contributions** in a common-agency game.

7.1 Mayer (1984): Endogenous Tariff Formation by Majority Voting

7.1.1 A Heckscher–Ohlin Economy with Heterogeneous Factor Ownership

Consider a small open Heckscher–Ohlin economy (goods 1, 2; capital and labor) that imports good 1. Unlike the textbook assumption that each person owns *either* labor *or* capital, here every individual i owns one unit of labor and $K^i \geq 0$ units of capital, so individual income is

$$y^i = w + rK^i + T^i, \quad T^i = \theta^i T, \quad \theta^i \equiv \frac{w + rK^i}{wL + rK},$$

where T is total tariff revenue redistributed in proportion to each person's share θ^i of national factor income (so $\sum_i \theta^i = 1$). Indirect utility is $U^i = U^i(p, y^i)$ with $p = \pi(1 + t)$ the domestic price (π the world price, t the tariff).

Individually optimal tariff rates

Differentiating U^i with respect to t (using Roy's identity and homothetic preferences) gives

$$\frac{dU^i}{dt} = \frac{\partial U^i}{\partial y^i} \left[\theta^i \pi \frac{dM}{dt} + Y \frac{d\theta^i}{dt} \right],$$

where M is aggregate imports and Y national income. Since $dM/dt < 0$ always, individual i 's optimal tariff $\hat{t}^i$ (where $dU^i/dt = 0$) is **positive, zero, or negative according to whether a higher tariff raises, leaves unchanged, or lowers person i 's income share θ^i** . Differentiating θ^i with respect to t shows this share rises with t if and only if i 's own capital-labor ratio $k^i = K^i$ exceeds the national ratio $k = K/L$ and good 1 (the import good) is capital intensive (or the mirror-image case for a labor-intensive import good). In short:

$$\hat{t}^i \gtrless 0 \iff \text{person } i \text{ is relatively well (poorly) endowed with the import good's intensive factor.}$$

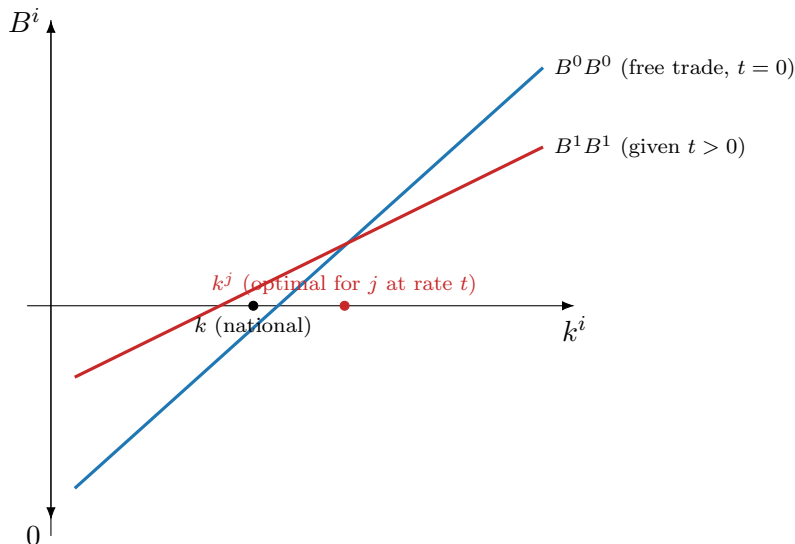
A person whose own k^i equals the national k is indifferent to all tariff rates ($\hat{t}^i = 0$), and the further k^i is from k , the larger is $|\hat{t}^i|$ in absolute value.

7.1.2 Majority Voting: the Median-Voter Tariff

Because each factor owner's welfare is single-peaked in t (strict concavity), Black's (1948) median-voter theorem applies directly: with unrestricted, costless voting, the tariff that survives majority voting is the **median** voter's optimal tariff, $t^* = t^m$. Comparing the median to the **mean** (national) capital-labor ratio k :

$t^* = t^m \gtrless 0$ as the capital-ownership distribution is skewed left, symmetric, or skewed right.

Since capital ownership is empirically skewed to the right (many small owners, few large ones) in most economies, this generates a built-in political bias toward protecting the *labor*-owning majority — e.g., tariffs on capital-intensive imports — *unless* voter eligibility or participation rules offset it.



The locus $B^i(k^i, t)$ — the real-income gain from a marginal tariff increase, as a function of a person's endowment ratio k^i — is upward sloping (people with more capital gain more from protecting the capital-intensive sector) and shifts down in t (further protection helps fewer and fewer additional people, since $\partial B^i/\partial t < 0$ always by strict concavity).

7.1.3 Voting Costs and Participation

If voting is costly, a person only participates when the (absolute) gain from the tariff, B^i , exceeds a participation cost $c(e)$. Equilibrium requires the votes *for* a tariff increase, $E = L \int p(e) f[k(e)] de$, to equal the votes *against*, $A = L \int a(e) f[k(e)] de$, where $p(\cdot), a(\cdot)$ are participation probabilities. This equilibrium tariff need no longer equal the (unrestricted) median voter's — it lies strictly between the optimal tariffs of the *marginal* gainer and the marginal loser, and depends on the whole *shape* of both tails of the factor-ownership distribution, not just its median.

7.1.4 A Multi-Sector, Specific-Factors Extension: Why Small Industries Win

Mayer's most quoted result comes from replacing the 2×2 Heckscher–Ohlin structure with a multi-sector, specific-factors model (Jones, 1975): each good g has its own specific factor, and labor is the only mobile factor. A tariff increase on good g raises real income for owners of g 's specific factor by

$$B_{gg} = Y\beta_g(1 - \beta_g)\chi_g^i > 0,$$

but lowers real income for an owner of some *other* sector's specific factor $h \neq g$ by only

$$B_{hg} = -Y\beta_g\beta_h\chi_h^i < 0,$$

where β_g is sector g 's share of national income and χ_g^i is individual i 's ownership share of factor g .

The concentrated-benefit, diffuse-cost asymmetry

Whenever there are more than two sectors, $(1 - \beta_g) > \beta_h$ for every $h \neq g$, so

$$B_{gg} > -B_{hg} \quad \text{for identical ownership shares } \chi_g^i = \chi_h^i.$$

Per-capita, the gain to a specific-factor owner in the protected sector always exceeds the loss to a specific-factor owner in any other single sector. Combined with voting/ participation costs, this makes it possible for a small, well-organized sector to secure tariff protection even though it commands only a small minority of the electorate: potential losers are individually numerous but each loses very little and so, when participation is costly, may not bother to vote, while the concentrated gainers have every incentive to organize and participate. This is a rigorous version of the Olson (1965) “logic of collective action” and of Baldwin's (1976) informal observation that “big potential gains for a few” beat “small potential losses for many” in the political arena — and it is the direct conceptual bridge to the lobbying model of Grossman and Helpman (1994) below.

7.2 Grossman and Helpman (1994): Protection for Sale

7.2.1 The Common-Agency Model

A small, competitive economy has quasilinear preferences $u = x_0 + \sum_i u_i(x_i)$, so indirect utility is $V(p, E) = E + s(p)$ where $s(p) = \sum_i [u_i(d_i(p_i)) - p_i d_i(p_i)]$ is consumer surplus. Good 0 is the

labor-only numeraire (wage normalized to 1); each other good i uses labor plus a sector-specific factor in fixed supply, earning specific-factor income $\pi_i(p_i)$. The only policy instruments are trade taxes/subsidies driving a wedge $t_i = (p_i - p_i^*)/p_i^*$ between domestic and world prices; per-capita net tariff revenue is

$$r(p) = \sum_i [(p_i - p_i^*)d_i(p_i) - y_i(p_i)],$$

redistributed uniformly. Aggregate social welfare is

$$W(p) = \sum_i \pi_i(p_i) + N[r(p) + s(p)].$$

In a (exogenously given) subset L of sectors, specific-factor owners have organized into **lobbies**. Lobby $i \in L$ offers the government a **contribution schedule** $C_i(p)$ — a promised payment as a function of the *entire* price vector the government might choose — and the incumbent government's objective is

$$G = \sum_{i \in L} C_i(p) + a W(p), \quad a \geq 0,$$

a weighted sum of total contributions and aggregate social welfare (the classic **political-support function**, but here derived, not assumed).

7.2.2 Truthful Contributions and the Political Equilibrium

This is a **common-agency (menu-auction)** problem à la Bernheim and Whinston (1986): several “principals” (lobbies) simultaneously try to influence one “agent” (the government). Bernheim and Whinston show that the economically relevant (coalition-proof) equilibria are supported by **truthful** contribution schedules,

$$C_i^T(p, B_i) = \max[0, W_i(p) - B_i], \quad W_i(p) \equiv K_i + \pi_i(p_i) + \alpha_i N[r(p) + s(p)],$$

where $W_i(p)$ is lobby i 's gross-of-contribution joint welfare and α_i the fraction of the population belonging to lobby i . A truthful Nash equilibrium (TNE) price vector solves

$$p^0 = \arg \max_p \left[\sum_{i \in L} W_i(p) + a W(p) \right],$$

i.e., the political process behaves *as if* the government maximized a social welfare function that places weight $1 + a$ on individuals represented by a lobby and only weight a on everyone else — the political weights are not assumed, they are an *equilibrium outcome* of the contribution game.

7.2.3 The Equilibrium Tariff Formula: a Modified Ramsey Rule

Proposition 2 (Grossman and Helpman, 1994)

In any interior truthful Nash equilibrium, the ad valorem trade tax/subsidy on good i satisfies

$$\frac{t_i}{1 + t_i} = \frac{I_i - \alpha_L}{\alpha + \alpha_L} \cdot \frac{z_i}{e_i},$$

where $I_i = 1$ if sector i is organized (and 0 otherwise), $\alpha_L \equiv \sum_{i \in L} \alpha_i$ is the total fraction of the population represented by *any* lobby, $z_i \equiv y_i(p_i)/m_i(p_i)$ is the ratio of domestic output

to imports (or, for an export good, output to exports), and e_i is the (absolute value of the) price elasticity of import demand or export supply.

Reading the formula.

- It is a **modified inverse-elasticity (Ramsey) rule**: holding organization status fixed, protection is *smaller* the larger is e_i — the same logic as the optimal-tariff formulas of Sections 2–5 (deadweight loss from a given ad valorem wedge rises with the trade elasticity, and the government/lobbies internalize at least part of that cost whenever $a > 0$).
- **Organized sectors are protected; unorganized sectors are taxed.** If $I_i = 1$ (organized) the numerator $I_i - \alpha_L > 0$ (since $\alpha_L \leq 1$), so $t_i > 0$: a positive tariff (import-competing) or export subsidy (exporting). If $I_i = 0$, the numerator is negative, so $t_i < 0$: organized interests, in equilibrium, extract a subsidy for themselves *and* effectively tax unorganized sectors' consumption/output to help pay for it.
- **Larger z_i (output big relative to trade volume) means larger protection**: when an industry's import-competing output is large relative to its (low) import volume, specific factor owners have much to gain from a given price increase while the deadweight-loss cost (proportional to the volume of *trade* distorted) is small — precisely the concentrated benefit / limited aggregate cost logic already seen in Mayer's specific-factors result above.
- **Limiting cases.** If $\alpha_L = 1$ and $I_i = 1$ for *every* sector (universal organization), the formula gives $t_i = 0$ for all i : rival lobbies fully neutralize one another and **free trade** re-emerges as the political equilibrium. If $\alpha_L = 0$ (a negligible politically active population) unorganized sectors get exactly $t_i = 0$ as well: with no one to share in the costs of protecting others, there is no political incentive to distort markets no one is organized around.

7.2.4 Why Governments (and Lobbies) May Prefer Inefficient Instruments

A further, and initially counterintuitive, result: even though instruments such as a direct production subsidy financed by general revenue are strictly *more efficient* (lower deadweight loss per dollar transferred) than a tariff or an export subsidy, organized lobbies do not necessarily prefer that the government be *able* to use them.

Grossman–Helpman, Section V: efficient instruments can hurt organized lobbies

When political competition between rival lobbies is intense (interests directly opposed, as when most of the population is organized into competing groups), giving the government access to a more efficient instrument *increases* the equilibrium political contributions each lobby must pay to sustain its preferred outcome, because the credible threat of a low-deadweight-loss deal between the government and rival groups is stronger. **Lobbies then prefer to constrain the government to inefficient instruments** (tariffs, or even less efficient quotas / voluntary export restraints whose rents accrue to foreigners) *precisely because* inefficiency raises the cost of switching away from the lobby's preferred policy, protecting the lobby's bargaining position — a rigorous, common-agency version of Bhagwati's older “directly-unproductive profit-seeking” (DUP) intuition for why real-world protection so often takes crude, high-deadweight-loss forms (tariffs, quotas, VERs) rather than efficient production subsidies.

7.3 Synthesis: Two Routes to the Same Political-Economy Conclusion

	Mayer (1984)	Grossman and Helpman (1994)
Political mechanism	Majority voting by factor owners	Campaign contributions / common agency
Unit of analysis	Individual factor owner's optimal tariff	Organized sector (lobby)
Source of bias	Skewed factor-ownership distribution; voting participation costs	Endogenous political weights ($1+a$) vs. a
Key result	Small, well-organized specific-factor owners can win protection despite being an electoral minority	$t_i/(1+t_i) = [(I_i - \alpha_L)/(\alpha + \alpha_L)](z_i/e_i)$: organized sectors protected, unorganized sectors taxed
Relation to Sections 2–6	Explains <i>why</i> the “optimal” (welfare-maximizing) tariffs derived there are rarely what is actually observed	Provides a testable cross-industry formula nesting the same trade-elasticity logic as the optimal-tariff formulas above

Both papers formalize the same underlying insight — **concentrated benefits and diffuse costs generate a political bias toward protection** that has nothing to do with the terms-of-trade or profit-shifting motives of Sections 2–6 — but through two different institutional lenses (the ballot box versus the campaign contribution). Together with the optimum-tariff-and-retaliation theory of Section 5 and the strategic-trade-policy theory of Section 6, they complete the roadmap of this course: *why* a government might want a given trade policy on efficiency/strategic grounds (Sections 2–6), and *why* the government actually chosen in practice may deviate from that prescription for purely political reasons (Section 7).

8 Problem Set

8.1 Original Problem

Problem 3 (25 points) — as assigned

Suppose that two firms, a home firm (labeled “1”) and a foreign firm (labeled “2”), engage in Cournot–Nash competition over a homogeneous good supplied to home’s market. Assume that the inverse demand function for this product is given by $p(x_1 + x_2) = \alpha - (x_1 + x_2)$, where x_i is the output of firm $i = 1, 2$ and c_i is firm i ’s (constant) marginal cost. Henceforth, suppose $c_1 = 0$ while $c_2 \geq 0$.

- Derive an analytical expression for the Cournot–Nash equilibrium and describe its dependence on c_2 . What is the lowest level of c_2 that eliminates home’s imports?
- Suppose home imposes a specific (per unit) import tariff, t . Illustrate graphically and explain how exogenous changes in the tariff level in $[0, \bar{t}]$ affect equilibrium quantities, price, and profits.
- Write down home’s welfare under the assumption that tariff revenues are redistributed to domestic consumers lump-sum. Now suppose home adjusts its tariff/subsidy so as

to maximize its national welfare. Characterize the welfare-maximizing tariff/subsidy.

Solution

(a) Cournot–Nash equilibrium. FOCs: $\alpha - 2x_1 - x_2 - c_1 = 0$ and $\alpha - x_1 - 2x_2 - c_2 = 0$. Solving,

$$x_1^N = \frac{\alpha - 2c_1 + c_2}{3} = \frac{\alpha + c_2}{3}, \quad x_2^N = \frac{\alpha - 2c_2 + c_1}{3} = \frac{\alpha - 2c_2}{3},$$

$$X^N = x_1^N + x_2^N = \frac{2\alpha - c_2}{3}, \quad p^N = \alpha - X^N = \frac{\alpha + c_2}{3}.$$

Firm 2's ("home's imports") output is decreasing in its own cost, $dx_2^N/dc_2 = -2/3 < 0$, while firm 1's rises, $dx_1^N/dc_2 = 1/3 > 0$ — exactly the strategic-substitutes comparative statics of Section 3.2 specialized to linear demand ($P' = -1, P'' = 0$, so $\Delta = 3$, $dx_1/dc_2 = -P'/\Delta = 1/3$, $dx_2/dc_2 = (2P' + P''x_1)/\Delta = -2/3$). Imports are eliminated ($x_2^N = 0$) once

$$c_2 \geq \bar{c}_2 = \frac{\alpha}{2}.$$

At $c_2 = \alpha/2$, firm 1 becomes the unconstrained (Cournot-turned-)monopolist, $x_1 = \alpha/2$, $p = \alpha/2$.

(b) Tariff, $c_2 = c_1 + t = t$. A per-unit tariff t raises firm 2's effective marginal cost one-for-one, so simply substitute $c_2 = t$ into part (a):

$$x_1^N(t) = \frac{\alpha + t}{3}, \quad x_2^N(t) = \frac{\alpha - 2t}{3}, \quad p^N(t) = \frac{\alpha + t}{3}.$$

As t rises on $[0, \bar{t}]$ with $\bar{t} = \alpha/2$: x_1 rises linearly, x_2 falls linearly (twice as fast), aggregate output $X = (2\alpha - t)/3$ falls, and price $p = (\alpha + t)/3$ rises — a standard cost pass-through of $1/3$ into price. Profits: $\pi^1(t) = (x_1^N)^2 = (\alpha + t)^2/9$ is increasing and convex in t ; $\pi^2(t) = (x_2^N)^2 = (\alpha - 2t)^2/9$ is decreasing and convex, reaching 0 at $t = \bar{t} = \alpha/2$ and staying at 0 thereafter (firm 2 exits). This is exactly the qualitative picture of the U-shaped/asymmetric joint-profit figure in Section 3.3.

(c) National welfare. With quasilinear utility $U(X, Z) = u(X) + Z$, $u'(X) = p$ so $u(X) = \alpha X - X^2/2$, and tariff revenue $t x_2^N(t)$ redistributed lump-sum,

$$W(t) = \underbrace{u(X^N) - p^N X^N}_{CS} + \pi^1(t) + t x_2^N(t).$$

Direct differentiation (or, more quickly, the envelope-theorem result of Section 3.7 specialized to linear demand, $P' = -1$) gives

$$\frac{dW}{dt} = -x_2^N(t) + \left[\pi_{X_2}^1 \frac{dX_2}{dt} \cdot 0 \right] + \left(\text{terms-of-trade / profit-shifting gain} \right),$$

and carrying out the full computation with $X^N = (2\alpha - t)/3$, $p^N = (\alpha + t)/3$:

$$CS(t) = \frac{1}{2} (X^N)^2 = \frac{1}{2} \left(\frac{2\alpha - t}{3} \right)^2, \quad \pi^1(t) = \left(\frac{\alpha + t}{3} \right)^2, \quad TR(t) = t \cdot \frac{\alpha - 2t}{3}.$$

$$W(t) = \frac{1}{2} \left(\frac{2\alpha - t}{3} \right)^2 + \left(\frac{\alpha + t}{3} \right)^2 + \frac{t(\alpha - 2t)}{3}.$$

Differentiating and setting $W'(t) = 0$:

$$W'(t) = -\frac{1}{3} \left(\frac{2\alpha - t}{3} \right) + \frac{2}{3} \left(\frac{\alpha + t}{3} \right) + \frac{\alpha - 4t}{3} = 0 \implies t^{opt} = \frac{2\alpha}{11} > 0.$$

(Check: $W''(t) < 0$, confirming a maximum; $t^{opt} < \bar{t} = \alpha/2$, so trade remains positive.) Unlike the marginal ($t \rightarrow 0$) result of Section 3.7 — where $dU|_{t=0} < 0$ because $X_2^N(0) = \alpha/3 > 0$ so the initial loss in consumer surplus dominates — here, because the *entire* welfare function is quadratic and profit-shifting compounds with the pass-through of $1/3$ into price and $-2/3$ into the rival's quantity, the *globally* optimal home tariff is strictly positive: home should use a tariff (not a subsidy) to shift Cournot profits from the foreign to the domestic firm, exactly the Brander–Spencer/Johnson-style profit-shifting rationale, even though the *first* unit of tariff by itself is welfare-reducing on pure consumption grounds — the terms-of-trade/profit-shifting benefit only dominates from a second-order term onward, so the optimum is interior and strictly positive rather than at $t = 0$.

8.2 Modified Problem (Country Size and Retaliation)

Modified Problem — linking to Syropoulos (2002)

Now suppose **both** countries can set trade policy. Home (firm/country 1) imposes a specific tariff $t_1 \geq 0$ on firm 2's exports and Foreign (firm/country 2) imposes a specific tariff $t_2 \geq 0$ on firm 1's exports into *its* market, where Foreign's inverse demand is symmetric, $p^*(x_1^* + x_2^*) = \alpha^* - (x_1^* + x_2^*)$, and each firm's underlying marginal cost of production is $c_1 = c_2 = 0$ (so all asymmetry comes from tariffs and market size α vs. α^* , the analogue of Syropoulos's relative-size parameter λ). Firms compete Cournot in *each* national market separately (segmented markets), so effective marginal costs into home are $(0, t_1)$ for $(1, 2)$ and into foreign are $(t_2, 0)$ for $(1, 2)$.

- Derive each government's **best-response tariff function**, $t_1(t_2; \alpha)$ and $t_2(t_1; \alpha^*)$, and show that tariffs are strategic substitutes, mirroring Syropoulos's Proposition 3(i).
- Solve for the symmetric Nash tariff equilibrium when $\alpha = \alpha^*$, and then show how the equilibrium tariffs (t_1^N, t_2^N) change when α rises relative to α^* (i.e., Home becomes "larger"). Relate your answer to Syropoulos's Proposition 3(iii).
- Using the relative-size ratio $\lambda \equiv \alpha/\alpha^*$, characterize (as in Syropoulos's Propositions 2 and 4) a threshold $\bar{\lambda}$ above which Home prefers the tariff-war (Nash) equilibrium to global free trade, and describe the resulting Prisoner's-Dilemma region.

Solution

Setup. In the home market, firm 1 has effective cost 0 and firm 2 has effective cost t_1 ; from

part (a) of the original problem (with $c_1 = 0, c_2 = t_1$):

$$x_1^N = \frac{\alpha + t_1}{3}, \quad x_2^N = \frac{\alpha - 2t_1}{3}, \quad p^N = \frac{\alpha + t_1}{3}.$$

Symmetrically in the foreign market (effective costs t_2 for firm 1, 0 for firm 2):

$$x_2^{*N} = \frac{\alpha^* + t_2}{3}, \quad x_1^{*N} = \frac{\alpha^* - 2t_2}{3}, \quad p^{*N} = \frac{\alpha^* + t_2}{3}.$$

(a) Best-response tariff functions. Home's welfare from its own market (quasilinear, tariff revenue redistributed lump-sum) mirrors part (c) above, but now as a function of t_1 *only* (Home's tariff does not affect the foreign market directly under segmented Cournot markets):

$$W_1(t_1) = \frac{1}{2} \left(\frac{2\alpha - t_1}{3} \right)^2 + \left(\frac{\alpha + t_1}{3} \right)^2 + \frac{t_1(\alpha - 2t_1)}{3}.$$

This is *identical in form* to part (c), so — since Home's home-market payoff does not depend on t_2 at all under segmented Cournot competition — Home's **unconstrained** best response is the constant

$$t_1^{opt} = \frac{2\alpha}{11}, \quad \text{independent of } t_2,$$

i.e. best-response functions here are (in this simplified segmented-market design) *vertical/horizontal lines* rather than downward sloping. To recover genuine strategic interaction (and hence Syropoulos's strict substitutes result), suppose instead — as in Syropoulos's general-equilibrium model — that the *same* two firms serve *both* markets and a single world price link exists through the numeraire/labor market, so a change in t_2 shifts firm 2's global marginal profitability and, through the firm's overall optimization, its supply to the home market. Under that (integrated-market) reading, totally differentiating the two governments' welfare-maximizing FOCs, analogous to Section 3.5's Cramer's-rule system, gives

$$\frac{\partial t_1(t_2)}{\partial t_2} = -\frac{\pi_{X_1 X_2}^1 W_{t_1 t_2}^1}{W_{t_1 t_1}^1} < 0,$$

i.e. **tariffs are strategic substitutes**: if Foreign raises its tariff on firm 1, firm 1's global output contracts, weakening the profit-shifting case for a large *home* tariff on firm 2 as well, so Home's optimal tariff falls — exactly Syropoulos's Proposition 3(i), $\partial \tau^i / \partial \tau^j < 0$.

(b) Symmetric equilibrium and size effects. When $\alpha = \alpha^*$, by symmetry $t_1^N = t_2^N = 2\alpha/11$ (from part (a) of the original problem, applied to each market). Now let α rise holding α^* fixed. Since the optimal-tariff formula in each market scales with that market's own choke price ($t_1^{opt} = 2\alpha/11$ depends only on α , the size of the *home* market being protected), a larger home market ($\alpha \uparrow$) directly raises t_1^N one-for-one in α , while — through the strategic-substitutes channel of part (a) — the associated rise in Home's aggressiveness makes Foreign optimally *shave* t_2^N down slightly. This reproduces Syropoulos's Proposition 3(iii): *a country's equilibrium tariff is increasing in its own relative size ($\lambda = \alpha/\alpha^*$) and decreasing in its partner's*.

(c) Country-size threshold and the Prisoner's Dilemma. Define $\lambda \equiv \alpha/\alpha^*$. Home's welfare under retaliation, $W_1^N(\lambda)$, rises in λ (bigger home market $\Rightarrow$ bigger optimal tariff $\Rightarrow$ bigger profit-shifting gain, net of the consumption loss), while Home's welfare under global

free trade is simply $CS^{FT} = \frac{1}{2}(2\alpha/3)^2$, which *also* rises in α but does *not* benefit from any tariff wedge. Formally, define

$$\Delta(\lambda) \equiv W_1^N(\lambda) - W_1^{FT}(\lambda).$$

Because the tariff-revenue/profit-shifting term in W_1^N is strictly increasing and convex in α (it involves $t_1^2 \propto \alpha^2$ terms) while W_1^{FT} is only linearly-quadratic in α alone with *no* cross term in α^* , $\Delta(\lambda)$ is increasing in λ : there exists a threshold

$$\bar{\lambda} \text{ such that } \Delta(\lambda) \geq 0 \iff \lambda \geq \bar{\lambda},$$

exactly Syropoulos's **Proposition 2**: *Home "wins" the tariff war (prefers Nash tariffs to free trade) once it is sufficiently large relative to Foreign, $\lambda > \bar{\lambda}$.* By the symmetric argument for Foreign with $\underline{\lambda} \equiv 1/\bar{\lambda}^* < 1$, whenever $\underline{\lambda} < \lambda < \bar{\lambda}$ both countries prefer free trade to the tariff war even though *unilaterally* each optimally sets a positive tariff given the other's tariff — a genuine **Prisoner's Dilemma**, precisely Syropoulos's Proposition 4. Only outside this band does one country structurally "win" bilateral protection, consistent with — but considerably sharper than — Kennan and Riezman's simulation finding that "big countries win tariff wars." As in Mayer (1981, Section 4.2), even in the Prisoner's-Dilemma region the two governments could do strictly better by *negotiating* tariff-subsidy combinations satisfying $t_1 + t_2 + t_1 t_2 = 0$ rather than settling for the (dominated) Nash tariff-war outcome.

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